

ODE Report — Automated Fish Length Estimation with Vision Foundation Models

ODE = Overview · Data · Execution — a structured, reproducible record of the machine-learning workflow, following the seminar's Scientific-Writing protocol.

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O — OVERVIEW

Research question. Can a modern image encoder or vision foundation model beat the reproduced AutoFish MobileNetV2 baseline at estimating fish length from a photo, under identical data, split, and metrics?

Target. Reproduce the paper's baseline (0.62 cm non-occluded MAE), then run a controlled encoder-swap comparison. Secondary tasks: species classification; a reliable improvement inside a foundation model.

Scope & assumptions. - Task is **regression** (predict a continuous length in cm), not detection/classification/segmentation. - Single fish per crop; bounding box provided by the dataset (we do not train a detector). - Only the **encoder** changes between length experiments; everything else is held fixed.

Why this setup. Reproducing the baseline first calibrates the whole pipeline, so any later difference is attributable to the encoder rather than to implementation error. A group-level split prevents the same fish leaking across train/test.

Contributions (current state). 1. Baseline reproduced to within 0.013 cm (non-occluded). 2. Controlled single-model comparison of 5 encoders (9 length runs). 3. A reliable, multi-seed improvement *within* DINOv2: patch-token pooling beats the CLS token. 4. Species classification across 4 encoders. 5. **EfficientNet-B0 reaches 0.781 cm — within 0.010 cm of the baseline** at a basic recipe; a validation-based recipe search is under way to try to cross it with a single model.

D — DATA

Dataset. AutoFish (Bengtson et al., WACV Workshops 2025), Hugging Face `vapaau/autofish`: 1,500 top-view RGB images, 454 unique fish, 25 groups, **18,157** instance annotations. Each annotation has a species label, a `fish_id`, a hand-measured `length` (cm), a bounding box, and a polygon segmentation.

Labels. Length in cm (regression target); species (7 classes: cod, haddock, hake, horse_mackerel, other, saithe, whiting) for the classification task.

Preprocessing (`scripts/`): 1. `build_autofish_index.py` → single `index.csv` (18,157 rows) joining annotations, images, species. 2. `make_crops.py` → per-annotation **masked, square** crops resized to 224×224 (background outside the fish polygon is blacked out; square crop preserves aspect ratio). 0 missing crops. 3. ImageNet normalization; ColorJitter augmentation on the training split only.

Splits (official, group-level). Train 15 groups / Validation 5 / Test 5. Test = **3,759** annotations (1,879 non-occluded [Set1+Set2] + 1,880 occluded [All]).

Leakage control. An audit found one fish (`fish_id 113`) crossing splits via a singleton duplicate annotation (id 3759, in a train group while its 40 other images are in a test group). It was removed (`exclusions.json`); the audit then confirms **zero** fish cross any split. This is why the non-occluded test count is 1,879, not 1,880.

E — EXECUTION

Model. Encoder → feature vector, concatenated with 4 normalized bbox values → MLP regression head (Linear+BatchNorm+ReLU blocks → 1 output). Only the encoder differs between length experiments.

Models & configurations tried (full-test MAE, cm):

Encoder	Type	Adaptation	Recipe (Adam)	Full-test MAE
MobileNetV2 (baseline)	CNN	full FT	1e-3, 200 ep, batch 32	0.771
EfficientNet-B0	CNN	full FT	1e-4, 100 ep, batch 16	0.781
ConvNeXt-Tiny	CNN	full FT	1e-4, 100 ep, batch 16	0.914
CLIP ViT-B/32	Transformer	last-block FT	1e-4, 100 ep	0.958
CLIP ViT-B/32	Transformer	frozen	1e-4, 100 ep	1.002
DINOv2 ViT-S/14 (patch)	Transformer	frozen, patch pooling	1e-4, 100 ep	1.261 ± 0.054 (3 seeds)
DINOv2 ViT-S/14 (CLS)	Transformer	last-block FT	1e-4, 100 ep	1.439
DINOv2 ViT-S/14 (CLS)	Transformer	frozen	1e-3, 100 ep	1.738
DINOv2 ViT-S/14 (CLS)	Transformer	full FT	enc 1e-5 / 1e-6	1.778 / 2.132

Training protocol. L1 loss, Adam; best checkpoint selected on **validation** MAE; test set used once. Fixed seed (42) per run; one JSON config per experiment; hardware NVIDIA RTX 5000 Ada (32 GB), Python 3.11. New trainer options added this round: **cosine LR schedule + weight decay**.

Metrics. MAE (primary, cm), plus RMSE, MAPE, bias, R^2 ; reported on full test, non-occluded, and occluded subsets. Baseline full-test: RMSE 1.268, MAPE 2.41%, bias +0.035, R^2 0.947.

Key results. - **Baseline reproduction:** 0.633 cm non-occluded vs paper 0.62 (Δ 0.013) → pipeline validated. - **Single-model ranking:** MobileNetV2 best; **EfficientNet-B0 within 0.010 cm**; supervised CNNs top the ranking; foundation models trail. - **Reliable DINOv2 improvement (3 seeds, identical settings):** patch-token pooling 1.261 ± 0.054 vs CLS token 1.843 ± 0.023 — a 0.58 cm gain with non-overlapping ranges. Last-block fine-tuning did not help (1.345). Best DINOv2 configuration = frozen patch pooling. - **Species classification (accuracy):** ConvNeXt 99.6%, MobileNetV2 99.1%, DINOv2 98.2%, CLIP 95.1% — all strong; DINOv2 rises from last (length) to near-top (species), evidence that foundation features suit semantics over precise geometry (hypothesis).

Validation-based recipe search (in progress). Because EfficientNet-B0 used a weaker recipe than the baseline, we run stronger recipes (cosine LR + weight decay + tuned LR + 200 epochs) for EfficientNet-B0 and ConvNeXt-Tiny, select the winner on **validation**, and report once on **test**. Persistent execution (`scripts/run_beat_baseline_queue.sh`) on the GPU server; results committed as they land.

Findings & interpretation. - Supervised CNNs (MobileNetV2, EfficientNet-B0, ConvNeXt) fill the top of the single-model ranking; general foundation features transfer only partially (CLIP) or poorly (DINOv2) for precise metric regression. Stated as an observation; mechanistic reasons are hypotheses. - EfficientNet-B0's near-baseline result at a basic recipe indicates the baseline is likely beatable by a single model with proper tuning.

Limitations. - Single training run for most single models (multi-seed done for the DINOv2 patch-vs-CLS study). - Foundation models tested at small scale (ViT-S/14, ViT-B/32) with limited fine-tuning budgets. - No single model beats the baseline yet; mask segmentation not yet evaluated. - The occluded-set reproduction difference vs the paper (0.909 vs 1.38) is unexplained.

Future work. Finish the recipe search (likely to cross the baseline); EfficientNet-B2; patch pooling for CLIP; larger ViT variants; multi-seed the top single models; mask segmentation as a third task; DINOv3 (pending weight access).

Reproducibility. Fixed seeds; official group split hard-coded; versioned configs; requirements.txt; per-run config/history/metrics/predictions; checkpoints on server; persistent queue scripts. Generative AI assisted with code scaffolding and drafting; all experiments and conclusions were verified by the authors.